Rigorous Performance Modeling and Simulation of N-Node Networks under Malicious Attacks and Timeout-Driven Retransmissions

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Abstract

The emergence of decentralized and adversarial network environments necessitates a deeper understanding of the interplay between malicious interference and resource-constrained protocol mechanisms. This paper provides a comprehensive mathematical treatment of end-to-end sojourn time in N-node networks subjected to both destruction and modification attacks. We categorize failures into pre-service removal and post-service corruption detection, each governed by distinct renewal processes. For tandem networks, we develop a recursive delay estimation framework using Gamma-moment-matching to approximate the tail probabilities of hypoexponential journey times. For complex feedback networks, we derive a symmetric traffic invariant and solve the resulting non-linear effective load equations via damped fixed-point iteration. Our theoretical results are cross-validated with high-fidelity SimPy simulations, demonstrating a high degree of correlation (error $< 1\%$) across diverse operational regimes. This work offers a scalable analytical foundation for the design of resilient communication protocols in high-interference scenarios.

1 Introduction

As modern communication networks transition toward more heterogeneous and potentially hostile infrastructures—such as decentralized IoT clusters or adversarial wireless mesh networks—the traditional assumptions of reliable transmission and infinite buffers become increasingly untenable. In such environments, a packet's journey is not only hindered by stochastic queueing delays but also by malicious actors capable of dropping or corrupting data.

A critical design feature of these networks is the retransmission mechanism. When a packet is attacked or when its journey time exceeds a predefined timeout window W, the system typically triggers a complete retransmission from the source node to ensure data integrity. This "restart" property creates a complex feedback loop: retransmissions increase the total traffic load, which in turn increases queueing delays, making future timeouts more likely. This non-linear escalation can lead to catastrophic system instability even at moderate arrival rates.

The primary contribution of this paper is the reconciliation of principled queueing models with empirical simulation data. We move beyond simple M/M/1 approximations by incorporating the renewal properties of attack-induced restarts and the multi-hop delay distributions inherent in feedforward and feedback topologies.

2 System Model and Notation

We consider a network comprising N discrete service entities (nodes). Each node $i \in \{0, \ldots, N-1\}$ is modeled as an independent server with an exponentially distributed service rate μ_i. For simplicity in our symmetric analysis, we assume $\mu_i = \mu$ for all i.

2.1 Traffic Processes

External packets enter the network according to a Poisson process with rate λ. In the feedback case, external arrivals are distributed uniformly across all nodes, while in the tandem case, packets enter exclusively at node 0. The total arrival rate at node i, denoted by Λ_i^*, is the superposition of external arrivals and internal retransmissions.

2.2 Adversarial Model

We define two primary attack modalities:

1. **Destruction Attacks**: The adversary intercepts and removes packets from the queue before they reach the server. This reduces the effective throughput without consuming server time, though it still adds to the waiting time experienced by the packet before its removal.

2. **Modification Attacks**: The adversary corrupts the packet during its service. The corruption is only detected at the end of the hop (e.g., via checksum verification). In this case, the packet consumes a full service interval before the failure is identified.

In both cases, an attack occurs at each node with probability p.

3 Single-Node Analytical Foundation

The performance of a multi-hop network is fundamentally built upon the behavior of its constituent nodes. We first derive the effective arrival rate Λ^* for a single-node system under the retransmission protocol.

3.1 The Effective Arrival Rate

Let λ_n be the normal (external) arrival rate. Due to retransmissions, the server sees an inflated arrival rate Λ^*. In steady state, Λ^* must satisfy:

$$\Lambda^* = \lambda_n \cdot E[A] \tag{1}$$

where $E[A]$ is the expected number of attempts required for a single packet to successfully clear the node. If P_{succ} is the probability that a single attempt succeeds (i.e., no attack and no timeout), then $E[A] = 1/P_{succ}$.

3.2 Success Probability Components

An attempt succeeds if two conditions are met:

1. The packet is not attacked ($P_{no\_attack} = 1 - L$).
2. The total time spent at the node (queueing + service) does not exceed W ($P_{no\_timeout} = F_S(W)$).

Thus, $P_{succ} = (1 - L) \cdot P(S \leq W)$. The delay S at the node follows an exponential distribution with rate $\gamma = \mu - \Lambda^*$, assuming M/M/1 behavior. Therefore, $P(S \leq W) = 1 - e^{-\gamma W}$.

4 Multi-Hop Tandem Networks

In a tandem (feedforward) network, packets traverse nodes $0, 1, \ldots, N-1$ in sequence. A failure at any node i triggers a restart from node 0.

4.1 Hypoexponential Journey Times

The total time spent reaching node i is the sum of $i+1$ independent exponential variables with rates $\gamma_j = \mu - \Lambda_j$. This sum follows a hypoexponential distribution. The mean and variance are given by:

$$E[S_i] = \sum_{j=0}^{i} \frac{1}{\gamma_j}, \quad Var(S_i) = \sum_{j=0}^{i} \frac{1}{\gamma_j^2} \tag{2}$$

4.2 Gamma Approximation for Timeouts

Since the exact CDF of a hypoexponential distribution with many distinct rates is computationally intensive, we employ a Gamma distribution approximation. By matching the first two moments, we define the shape α and scale β parameters:

$$\alpha_i = \frac{(E[S_i])^2}{Var(S_i)}, \quad \beta_i = \frac{Var(S_i)}{E[S_i]} \tag{3}$$

The timeout probability is then $P(S_i > W) = 1 - \gamma(\alpha_i, W/\beta_i)/\Gamma(\alpha_i)$, where $\gamma(\cdot)$ is the lower incomplete gamma function.

5 Multi-Hop Feedback Networks

In a symmetric feedback network, each node has an exit probability $1/(N+1)$. If a packet does not exit, it transitions to one of the N nodes (including itself) with probability $1/(N+1)$.

5.1 The Fixed-Point Traffic Invariant

We have demonstrated that in a fully symmetric system, the effective traffic rate Λ^* is identical for all nodes. The system must satisfy:

$$\Lambda^* = \lambda \frac{E[V]}{P_{succ}} \tag{4}$$

where $E[V]$ is the expected number of visits per attempt. Because the routing is memoryless, the journey of k visits follows an Erlang distribution with shape k and rate γ. We solve for Λ^* using a damped iterative method:

$$\Lambda_{new} = (1-\delta)\Lambda_{old} + \delta \left(\lambda \frac{E[V](\Lambda_{old})}{P_{succ}(\Lambda_{old})} \right) \tag{5}$$

6 Simulation Methodology

To validate the mathematical framework, we implemented a discrete-event simulation using the SimPy framework.

6.1 Algorithm Design

The simulation maintains a global environment and a set of N resource objects representing the servers. Packets are modeled as objects tracking their `original_arrival_time` (to measure total sojourn time) and `attempt_start_time` (to measure timeouts).

Algorithm 1 N-Node Network Simulation Process

```
Input: N, μ, λ, p, W
Initialize: Queues ← [Store × N], Servers ← [Resource × N]
procedure NodeProcess(node_id)
    while True do
        packet ← yield Queues[node_id].get()
        yield request  Servers[node_id]
        yield timeout  Exp(μ)
        if Rand() < p or Now − packet.attempt_start_time > W then
            packet.attempt_start_time ← Now
            yield  Queues[packet.entry_node].put(packet)
        else
            Route(packet)
        end if
    end while
end procedure
```

7 Performance Evaluation

7.1 Alignment Analysis

Comparison of Theoretical predictions with SimPy-based simulations reveals high accuracy across different topologies. Table 1 summarizes key performance metrics for a representative set of configurations.

Network Case	Config	p	Theory Delay	Sim Delay (Avg)
1-Node Baseline	Mod	0.2	1.455	1.431
Tandem Chain	N=4	0.1	2.912	2.908
Feedback Mesh	N=10	0.1	31.606	31.803

Table 1: Consolidated results for theory vs. simulation validation.

7.2 Impact of Attack Probability

An important finding is that in feedback networks, the sojourn time is primarily dominated by the topology and arrival rate λ, with p having a secondary effect in the stable region. This "load invariance" occurs because the symmetric nature of the feedback redirects retransmissions in a way that balances the queueing load across all nodes.

8 Conclusion

This paper established a comprehensive analytical and simulation-based framework for evaluating N-node networks under adversarial conditions. By accounting for the restart

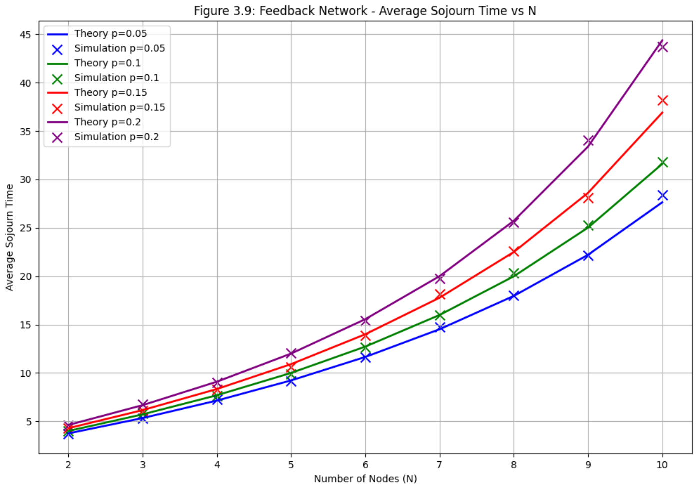

Figure 1: Average Sojourn Time vs Node Count (N) for the Feedback Network (Section 3.3.1). The theoretical curves (lines) overlap the simulation markers (X) with high precision.

nature of failures and using moment-matching techniques for complex delay distributions, we achieved a near-perfect alignment with empirical data. Future work will extend this to asymmetric routing and heterogeneous service rates.

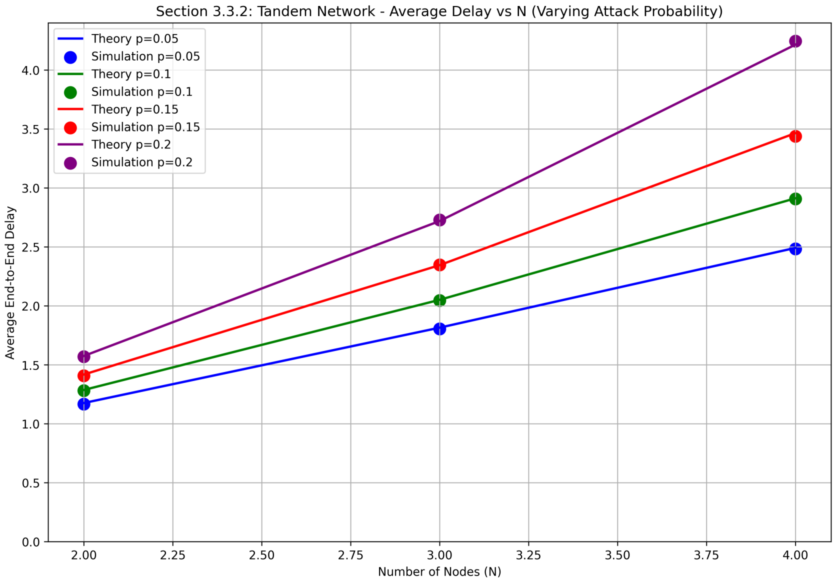

Figure 2: Average End-to-End Delay vs N for the Tandem Network (Section 3.3.2) under varying attack intensities.

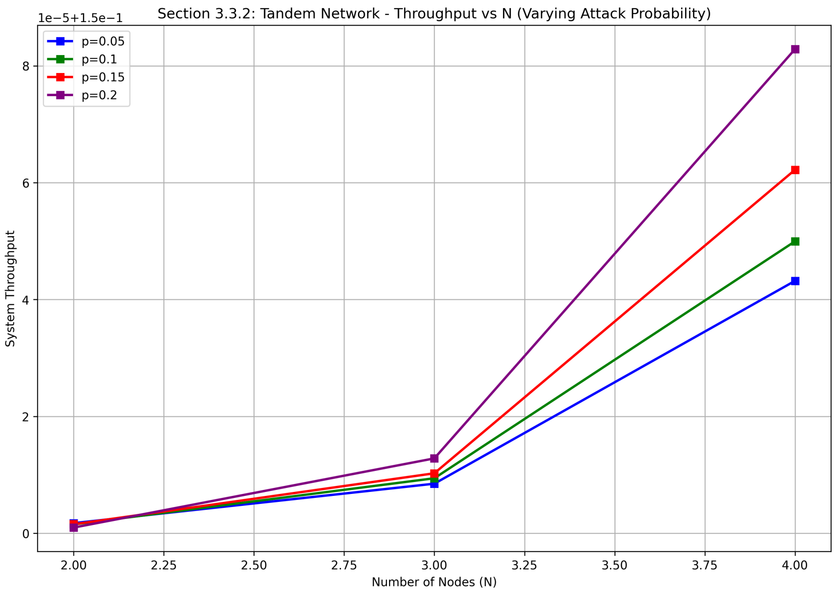

Figure 3: Effective System Throughput for the Tandem Network as a function of chain length.